

Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

Week-1

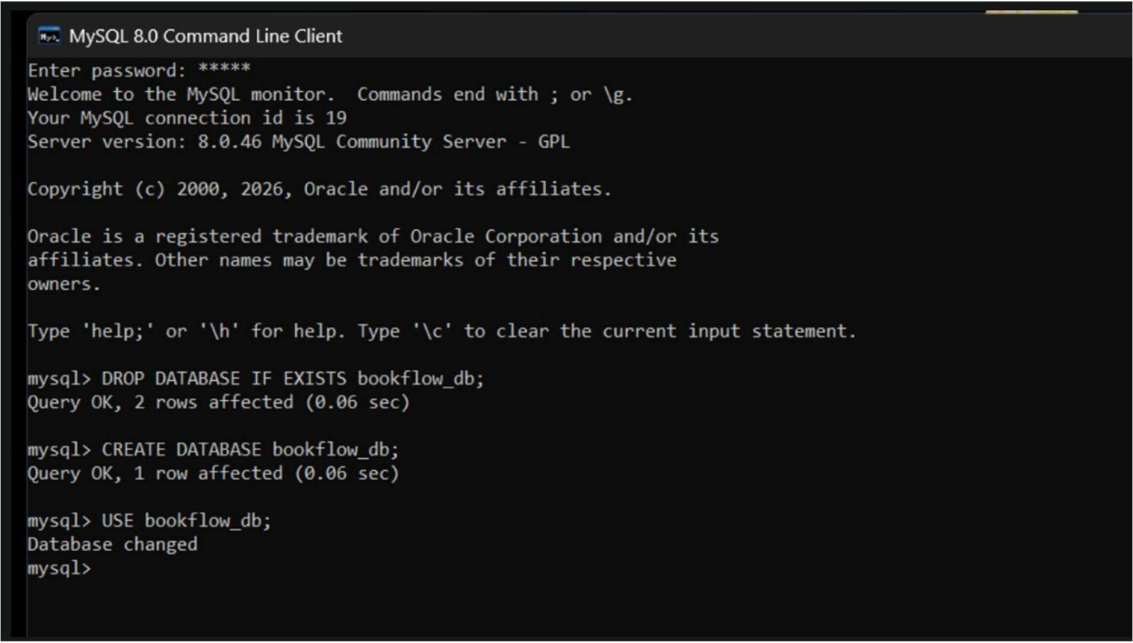
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Step 1 :Database Setup Query 1: Create the database

```
CREATE DATABASE IF NOT EXISTS bookflow_db;
```

Explanation: CREATE DATABASE creates a new, empty database (a container for tables). The IF NOT EXISTS clause makes the command safe to re-run: if bookflow_db already exists, MySQL skips creation instead of throwing an error. This is a DDL (Data Definition Language) command.

A screenshot of the MySQL 8.0 Command Line Client window. The window title is "MySQL 8.0 Command Line Client". The prompt "Enter password: *****" is shown. Below it, a welcome message says "Welcome to the MySQL monitor. Commands end with ; or \g. Your MySQL connection id is 19. Server version: 8.0.46 MySQL Community Server - GPL. Copyright (c) 2000, 2026, Oracle and/or its affiliates. Oracle is a registered trademark of Oracle Corporation and/or its affiliates. Other names may be trademarks of their respective owners. Type 'help;' or '\h' for help. Type '\c' to clear the current input statement." The user enters the command "mysql> DROP DATABASE IF EXISTS bookflow_db;" and the output is "Query OK, 2 rows affected (0.06 sec)". Then the user enters "mysql> CREATE DATABASE bookflow_db;" and the output is "Query OK, 1 row affected (0.06 sec)". Finally, the user enters "mysql> USE bookflow_db;" and the output is "Database changed". The prompt "mysql>" is shown at the bottom.

```
MySQL 8.0 Command Line Client
Enter password: *****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 19
Server version: 8.0.46 MySQL Community Server - GPL

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> DROP DATABASE IF EXISTS bookflow_db;
Query OK, 2 rows affected (0.06 sec)

mysql> CREATE DATABASE bookflow_db;
Query OK, 1 row affected (0.06 sec)

mysql> USE bookflow_db;
Database changed
mysql>
```

Query 2: Select the database

```
USE bookflow_db;
```

Explanation: USE switches the current session to bookflow_db so that every subsequent CREATE TABLE, INSERT, and SELECT runs inside this database. Without this step, the tables would be created in whatever database was previously active.

```
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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> DROP DATABASE IF EXISTS bookflow_db;
Query OK, 2 rows affected (0.06 sec)

mysql> CREATE DATABASE bookflow_db;
Query OK, 1 row affected (0.06 sec)

mysql> USE bookflow_db;
Database changed
mysql> CREATE TABLE books (
  ->   book_id INT AUTO_INCREMENT PRIMARY KEY,
  ->   title VARCHAR(255) NOT NULL,
  ->   isbn VARCHAR(13) NOT NULL UNIQUE,
  ->   published_year INT,
  ->   CONSTRAINT chk_published_year
  ->   CHECK (published_year < 2027)
  -> );
Query OK, 0 rows affected (0.06 sec)

mysql>
```

3. Step 2 Table Creation (DDL) with Constraints

Query 3: Create the books table

```
CREATE TABLE books (
  book_id    INT AUTO_INCREMENT PRIMARY KEY,
  title      VARCHAR(255) NOT NULL,
  isbn       VARCHAR(13) NOT NULL UNIQUE,
  published_year INT,
  CONSTRAINT chk_published_year CHECK (published_year < 2027)
);
```

Explanation: Line-by-line breakdown of every column and constraint:

- `book_id INT AUTO_INCREMENT PRIMARY KEY` — the PRIMARY KEY uniquely identifies each row; it is automatically NOT NULL and UNIQUE. AUTO_INCREMENT makes MySQL generate 1, 2, 3, ... automatically, so we never insert the ID manually.
- `title VARCHAR(255) NOT NULL` — NOT NULL means a book cannot be saved without a name. An INSERT with a NULL title is rejected.
- `isbn VARCHAR(13) NOT NULL UNIQUE` — UNIQUE guarantees no two books share the same ISBN. VARCHAR is used instead of INT because ISBNs can have leading zeros and ISBN-10 can end in the letter X.
- `published_year INT` with CHECK (`published_year < 2027`) — the named CHECK constraint (`chk_published_year`) rejects any year of 2027 or later, preventing "future" books from entering the system.

```
MySQL 8.0 Command Line Client
Your MySQL connection id is 19
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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> DROP DATABASE IF EXISTS bookflow_db;
Query OK, 2 rows affected (0.06 sec)

mysql> CREATE DATABASE bookflow_db;
Query OK, 1 row affected (0.06 sec)

mysql> USE bookflow_db;
Database changed
mysql> CREATE TABLE books (
  ->   book_id INT AUTO_INCREMENT PRIMARY KEY,
  ->   title VARCHAR(255) NOT NULL,
  ->   isbn VARCHAR(13) NOT NULL UNIQUE,
  ->   published_year INT,
  ->   CONSTRAINT chk_published_year
  ->   CHECK (published_year < 2027)
  -> );
Query OK, 0 rows affected (0.06 sec)

mysql>
```

Query 4: Create the members table

```
CREATE TABLE members (
  member_id INT AUTO_INCREMENT PRIMARY KEY,
  full_name VARCHAR(100) NOT NULL,
  email VARCHAR(150) NOT NULL UNIQUE
);
```

Explanation: • `member_id INT AUTO_INCREMENT PRIMARY KEY` — unique identity for every member, generated automatically. • `full_name VARCHAR(100) NOT NULL` — a member must have a name on record. • `email VARCHAR(150) NOT NULL UNIQUE` — NOT NULL means no signup without an email; UNIQUE means one account per email address. Together these satisfy the "no member can sign up without a valid email" rule.

```
MySQL 8.0 Command Line Client
-> published_year INT,
-> CONSTRAINT chk_published_year
-> CHECK (published_year < 2027)
-> );
Query OK, 0 rows affected (0.06 sec)

mysql> CREATE TABLE members (
-> member_id INT AUTO_INCREMENT PRIMARY KEY,
-> full_name VARCHAR(100) NOT NULL,
-> email VARCHAR(150) NOT NULL UNIQUE
-> );
Query OK, 0 rows affected (0.12 sec)

mysql> DESCRIBE books;
+-----+-----+-----+-----+-----+-----+
| Field      | Type      | Null | Key | Default | Extra      |
+-----+-----+-----+-----+-----+-----+
| book_id    | int       | NO   | PRI | NULL    | auto_increment |
| title      | varchar(255) | NO   |     | NULL    |              |
| isbn       | varchar(13) | NO   | UNI | NULL    |              |
| published_year | int       | YES  |     | NULL    |              |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.05 sec)

mysql>
```

Query 5 (verification): Inspect the table structures

Explanation: DESCRIBE shows the schema. Key = PRI confirms the PRIMARY KEY; Key = UNI confirms the UNIQUE constraint on isbn and email; Null = NO confirms NOT NULL columns. This is how an examiner verifies the constraints exist.

```
MySQL 8.0 Command Line Client
-> );
Query OK, 0 rows affected (0.12 sec)

mysql> DESCRIBE books;
+-----+-----+-----+-----+-----+-----+
| Field      | Type      | Null | Key | Default | Extra      |
+-----+-----+-----+-----+-----+-----+
| book_id    | int       | NO   | PRI | NULL    | auto_increment |
| title      | varchar(255) | NO   |     | NULL    |              |
| isbn       | varchar(13) | NO   | UNI | NULL    |              |
| published_year | int       | YES  |     | NULL    |              |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.05 sec)

mysql> DESCRIBE members;
+-----+-----+-----+-----+-----+-----+
| Field      | Type      | Null | Key | Default | Extra      |
+-----+-----+-----+-----+-----+-----+
| member_id  | int       | NO   | PRI | NULL    | auto_increment |
| full_name  | varchar(100) | NO   |     | NULL    |              |
| email      | varchar(150) | NO   | UNI | NULL    |              |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

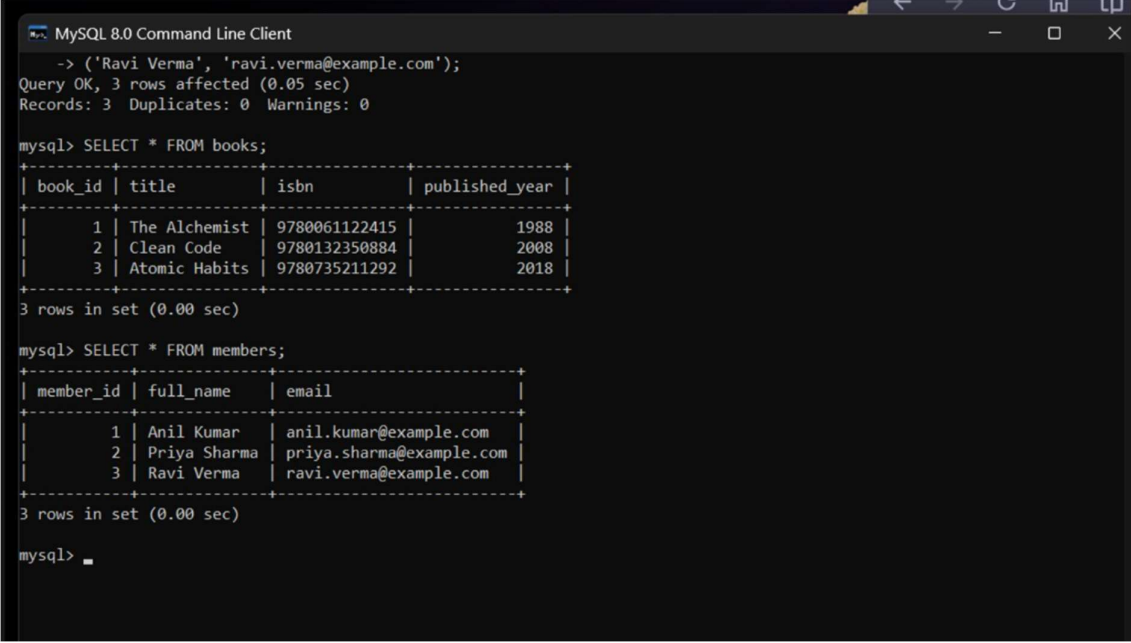
mysql>
```

Query 6: Insert 3 books

```
INSERT INTO books (title, isbn, published_year) VALUES
('The Alchemist', '9780061122415', 1988),
('Clean Code', '9780132350884', 2008),
('Atomic Habits', '9780735211292', 2018);
```

Query 7: Insert 3 members

```
INSERT INTO members (full_name, email) VALUES
('Anil Kumar', 'anil.kumar@example.com'),
('Priya Sharma', 'priya.sharma@example.com'),
('Ravi Verma', 'ravi.verma@example.com');
```



```
MySQL 8.0 Command Line Client
-> ('Ravi Verma', 'ravi.verma@example.com');
Query OK, 3 rows affected (0.05 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM books;
+-----+-----+-----+-----+
| book_id | title       | isbn          | published_year |
+-----+-----+-----+-----+
| 1       | The Alchemist | 9780061122415 | 1988           |
| 2       | Clean Code   | 9780132350884 | 2008           |
| 3       | Atomic Habits | 9780735211292 | 2018           |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> SELECT * FROM members;
+-----+-----+-----+
| member_id | full_name | email              |
+-----+-----+-----+
| 1         | Anil Kumar | anil.kumar@example.com |
| 2         | Priya Sharma | priya.sharma@example.com |
| 3         | Ravi Verma | ravi.verma@example.com |
+-----+-----+-----+
3 rows in set (0.00 sec)

mysql>
```

Test 1: Duplicate ISBN (UNIQUE violation)

```
INSERT INTO books (title, isbn, published_year)
VALUES ('Fake Copy', '9780061122415', 2000);
```

Output:

Explanation: The ISBN 9780061122415 already belongs to 'The Alchemist'. The UNIQUE constraint on isbn rejects the row, so two physical records can never claim the same ISBN.

Test 2: NULL title (NOT NULL violation)

```
INSERT INTO books (title, isbn, published_year)
VALUES (NULL, '9999999999999', 2010);
```

Output:

Explanation: The NOT NULL constraint fires because a book must have a name. The database refuses to store a nameless book, exactly as the requirement demands.

Test 3: Future publication year (CHECK violation)

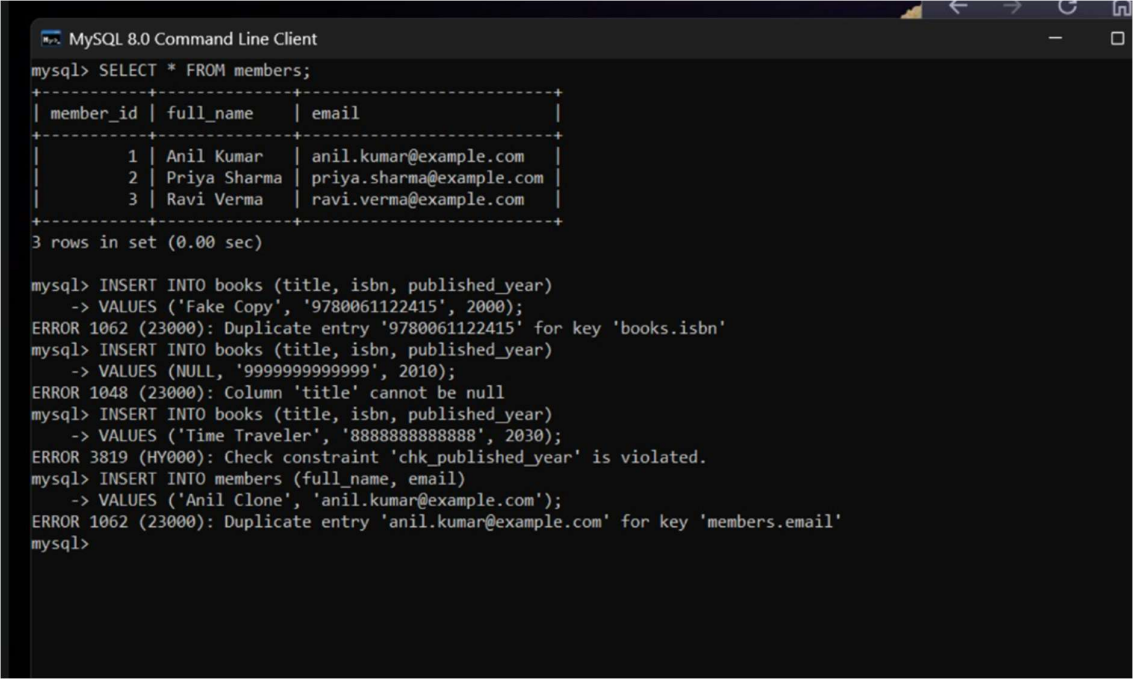
```
INSERT INTO books (title, isbn, published_year)
VALUES ('Time Traveler', '88888888888888', 2030);
```

Explanation: 2030 fails the condition `published_year < 2027`, so the named CHECK constraint `chk_published_year` blocks the insert. Garbage dates cannot enter the table.

Test 4: Duplicate email (UNIQUE violation)

```
INSERT INTO members (full_name, email)
VALUES ('Anil Clone', 'anil.kumar@example.com');
```

Explanation: `anil.kumar@example.com` is already registered to member_id 1, so the UNIQUE constraint on email rejects the second signup. One email = one account



```
MySQL 8.0 Command Line Client
mysql> SELECT * FROM members;
+-----+-----+-----+
| member_id | full_name | email |
+-----+-----+-----+
| 1 | Anil Kumar | anil.kumar@example.com |
| 2 | Priya Sharma | priya.sharma@example.com |
| 3 | Ravi Verma | ravi.verma@example.com |
+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> INSERT INTO books (title, isbn, published_year)
-> VALUES ('Fake Copy', '9780061122415', 2000);
ERROR 1062 (23000): Duplicate entry '9780061122415' for key 'books.isbn'
mysql> INSERT INTO books (title, isbn, published_year)
-> VALUES (NULL, '99999999999999', 2010);
ERROR 1048 (23000): Column 'title' cannot be null
mysql> INSERT INTO books (title, isbn, published_year)
-> VALUES ('Time Traveler', '88888888888888', 2030);
ERROR 3819 (HY000): Check constraint 'chk_published_year' is violated.
mysql> INSERT INTO members (full_name, email)
-> VALUES ('Anil Clone', 'anil.kumar@example.com');
ERROR 1062 (23000): Duplicate entry 'anil.kumar@example.com' for key 'members.email'
mysql>
```

```
MySQL 8.0 Command Line Client
mysql> INSERT INTO members (full_name, email)
-> VALUES ('Anil Clone', 'anil.kumar@example.com');
ERROR 1062 (23000): Duplicate entry 'anil.kumar@example.com' for key 'members.email'
mysql> DESCRIBE books;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| book_id | int | NO | PRI | NULL | auto_increment |
| title | varchar(255) | NO | | NULL | |
| isbn | varchar(13) | NO | UNI | NULL | |
| published_year | int | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> DESCRIBE members;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| member_id | int | NO | PRI | NULL | auto_increment |
| full_name | varchar(100) | NO | | NULL | |
| email | varchar(150) | NO | UNI | NULL | |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql>
```